Technical Report: Design, Layout, and Verification of a 400 V SiC MOSFET Isolated Half-Bridge Gate Driver PCB

1. Project Overview & Circuit Architecture

The objective of this project is to design, lay out, and simulate a 400 V DC half-bridge switching stage using Silicon Carbide (SiC) power MOSFETs. In power electronics applications such as DC-AC inverters and DC-DC converters, a microcontroller outputs low-voltage (5 V) pulse-width modulation (PWM) signals that cannot directly drive power switches operating at hundreds of volts.

This circuit bridges that gap using the **Texas Instruments UCC21520DW** isolated dual-channel gate driver (U1) to drive two **Wolfspeed C3M0120090D** 900 V SiC MOSFETs (Q1 and Q2).

1.1 Technical Background: Isolation and High-Side Drive Requirements

Operating a phase-leg power stage under a 400 V DC bus imposes two fundamental design constraints:

1. **Galvanic Isolation:** High-voltage switching transients and ground potential shifts must be physically separated from sensitive low-voltage microcontrollers. The capacitive isolation barrier inside the UCC21520DW provides 5.7 kVrms isolation between the primary control domain (GND) and the secondary power stage (GNDPWR, SW_NODE). This prevents high-voltage fault propagation, breaks ground loops, and protects digital logic from common-mode transients (CMTI).

2. **Floating High-Side Potential:** An N-channel power MOSFET requires a positive gate-to-source bias ($V_{GS} \approx 14$ V to 15 V) to conduct. While the low-side switch (Q2) has its Source referenced to a fixed potential (GNDPWR = 0 V), the high-side switch (Q1) has its Source tied to the switching node (SW_NODE). Because SW_NODE alternates between 0 V and 400 V during operation, the high-side gate-drive supply (VDDA relative to VSSA) must float on top of the switching potential to maintain enough gate voltage without referencing earth or system ground.

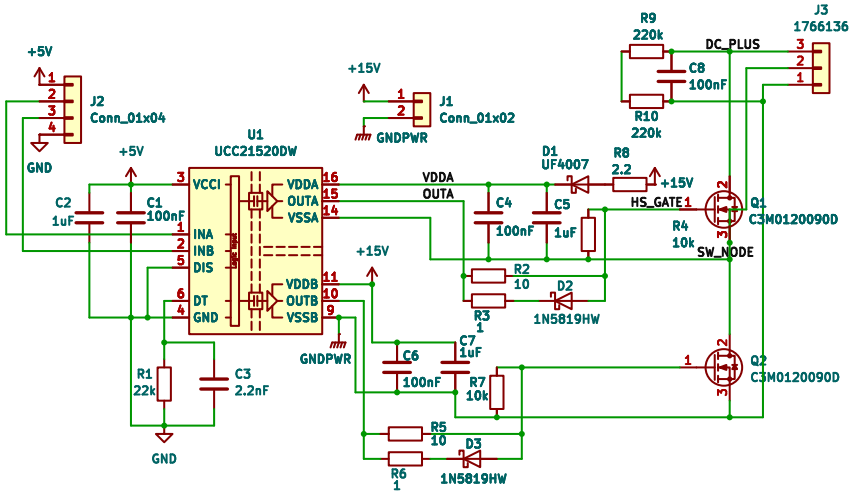


Figure 1 — Complete schematic of the 400 V SiC half-bridge gate driver in KiCad, showing the primary 5 V logic inputs (J2), the isolated gate driver (U1), the bootstrap charging circuit (D1, R8, C4, C5), asymmetric split gate resistors, and the 400 V power stage (Q1, Q2, C8, R9, R10).

Bill of Materials & Component Selection

Every component in the power stage was selected and derated to handle a 400 V DC bus safely, with orderable Manufacturer Part Numbers (MPNs) matched to their physical PCB footprints:

Ref	Qty	Value / Role	Footprint	Manufacturer	Part Number (MPN)	Selection Reasoning
U1	1	Isolated Gate Driver	SOIC-16W	Texas Instruments	UCC21520DW	Wide-body package gives >8 mm creepage between primary and secondary pins; 4 A source / 6 A sink peak current.
Q1, Q2	2	900V SiC MOSFET	TO-247-3 Vertical	Wolfspeed	C3M0120090D	900 V rating provides >2x safety margin above the 400 V DC bus; low R_DS(on) = 120 mΩ.
D1	1	Bootstrap Diode	DO-41 Axial	Generic	UF4007	Must block the full 400 V bus when Q1 is ON. Rated for 1000 V with ultrafast 75 ns reverse recovery.
D2, D3	2	Gate Turn-Off Diode	SOD-123 SMD	Generic	1N5819HW	40 V, 1 A Schottky diode with low forward voltage drop to create a fast turn-off path for the MOSFET gates.
C8	1	100nF Snubber Cap	Radial (P=15mm)	WIMA	MKP1J031004B00KSSD	630 V DC polypropylene film capacitor. Chosen over a 1000 V part because its slim 5 mm width fits the footprint without hitting R9.
R9, R10	2	220kΩ Bleeder	0207 Axial	Yageo	MFR-25FTE52-220K	Placed in series across the 400 V bus so each resistor only sees 200 V, staying safely below its 250 V working rating.

R1	1	22kΩ Dead-Time	0805 SMD	Yageo	RC0805FR-0722KL	Programs hardware dead-time on Pin 6 (DT) to ~220 ns.
R2, R5	2	10Ω Turn-On	0805 SMD	Yageo	RC0805FR-0710RL	Controls gate charging speed during MOSFET turn-on to limit di/dt and voltage ringing.
R3, R6	2	1Ω Turn-Off	0805 SMD	Yageo	RC0805FR-071RL	Provides a low-impedance discharge path during turn-off to clamp the gate against Miller bounce.
R4, R7	2	10kΩ Gate Pull-Down	0805 SMD	Yageo	RC0805FR-0710KL	Keeps MOSFET gates tied to Source (0 V) if the gate driver is unpowered.
R8	1	2.2Ω Bootstrap	0805 SMD	Yageo	RC0805FR-072R2L	Limits initial inrush current when charging the bootstrap capacitor at startup.
C1, C4, C6	3	100nF Decoupling	0805 SMD	Murata	GRM21BR71H104KA01L	50 V X7R ceramic capacitors placed close to IC supply pins (VCCI, VDDA, VDDB) for high-frequency decoupling.
C2, C5, C7	3	1µF Bulk Supply	0805 SMD	Murata	GRM21BR71H105KA12L	50 V X7R ceramic bulk energy storage for the primary, high-side bootstrap, and low-side supplies.
C3	1	2.2nF DT Filter	0805 SMD	Murata	GRM21BR71H222KA01L	Noise filter capacitor across the dead-time resistor R1 per TI datasheet guidelines.
J3	1	HV Terminal Header	1x03 (P=7.62mm)	Phoenix Contact	1766246	630 V rated GMSTBA header for DC_PLUS, SW_NODE, and GNDPWR (mates with plug 1767012).
J1, J2	2	Low-Voltage Headers	2.54mm Pitch	Generic	1x02 & 1x04 Male	Standard pin headers for the +15V gate supply (J1) and +5V / PWM logic signals (J2).

2. Subsystem Design & Hand Calculations

2.1 Dead-Time Programming (R1 and C3)

In a half-bridge, if both Q1 and Q2 conduct at the same time, they create a direct short-circuit across the 400 V DC supply (shoot-through). To prevent this, the UCC21520DW includes a programmable dead-time pin (DT, Pin 6) that enforces a brief delay between turning one switch OFF and turning the other switch ON.

According to the TI UCC21520DW datasheet, the dead-time (t_DT) is set by the resistor connected from Pin 6 to GND using the linear relation:
$t_{DT} \approx 10\ \text{ns/k}\Omega \times R1$

Using a standard R1 = 22 kΩ resistor:

t_DT ≈ 10 ns/kΩ × 22 kΩ = 220 ns

A 2.2 nF capacitor (C3) is placed in parallel with R1 as required by the datasheet to stabilize the pin voltage against switching noise. A generous 220 ns delay was chosen to ensure a wide safety margin against shoot-through for a first hardware prototype.

2.2 Bootstrap Supply Network (D1, R8, C4, C5)

Instead of using a separate isolated DC-DC converter to power the high-side driver (VDDA to VSSA), a bootstrap circuit is used. Whenever the low-side MOSFET (Q2) is turned ON, SW_NODE is pulled down to 0 V (GNDPWR). This allows the +15 V supply to charge the parallel bootstrap capacitors (C5 = 1 µF and C4 = 100 nF, total C_boot = 1.1 µF) through resistor R8 (2.2 Ω) and diode D1 (UF4007). When Q2 turns OFF and Q1 turns ON, SW_NODE rises to 400 V, D1 becomes reverse-biased and blocks the 400 V, and C4/C5 act as a floating battery for the high-side driver.

1. **High-Side Gate Drive Voltage (V_boot):** Because the charging current passes through the UF4007 diode (D1), which has a forward voltage drop of V_F ≈ 1.0 V, the voltage stored across the bootstrap capacitors is:
   V_boot = V_CC - V_F ≈ 15 V - 1.0 V = 14 V

2. **Voltage Droop When Turning On Q1 (ΔV_boot):** From the Wolfspeed C3M0120090D datasheet, the total gate charge required to switch the MOSFET on at 15 V is Q_g = 17.3 nC. Using the basic capacitor charge equation ΔV = ΔQ / C, when C4 and C5 (1.1 µF total) supply 17.3 nC to the gate, their voltage drops by:
   ΔV_boot = Q_g / C_boot = (17.3 × 10^-9 C) / (1.1 × 10^-6 F) ≈ 0.0157 V (or 15.7 mV)

   A droop of only 15.7 mV out of 14 V (~0.11%) proves that the 1 µF + 100 nF capacitor combination easily holds the high-side supply steady during switching.

3. **Startup Inrush Current Limit (R8):** If the bootstrap capacitors are completely empty when power is first applied, R8 (2.2 Ω) limits the peak charging current from the +15 V rail:
   I_peak = (15 V - 1.0 V) / 2.2 Ω ≈ 6.36 A
   This brief microsecond current pulse is well below the 30 A peak surge rating of the UF4007 diode.

2.3 Split Gate Resistor Network (R2/R3/D2 and R5/R6/D3)

A MOSFET gate behaves like a capacitor (C_iss) that must be charged to turn the switch ON and discharged to turn it OFF. Instead of using a single resistor for both directions, an asymmetric split network with a Schottky diode (1N5819HW) is used:

- **Controlled Turn-On (10 Ω):** When the driver outputs a high voltage (+14 V or +15 V), diodes D2 and D3 are reverse-biased (blocking). All gate charging current must flow through the 10 Ω resistors (R2 and R5). Slowing down the turn-on slightly limits the rate of current change (di/dt), which prevents excessive voltage spikes and ringing across the power loop.
- **Fast Turn-Off (1 Ω || 10 Ω = 0.91 Ω):** When the driver pulls its output to 0 V to turn the MOSFET OFF, current flows out of the gate through the forward-biased Schottky diode (D2/D3) and the 1 Ω resistor (R3/R6) in parallel with the 10 Ω resistor:
- $R_{off} = (1\ \Omega \times 10\ \Omega) / (1\ \Omega + 10\ \Omega) \approx 0.91\ \Omega$
- **Why fast turn-off matters (The Miller Effect):** When Q1 turns ON rapidly, SW_NODE jumps from 0 V to 400 V in nanoseconds (high dv/dt). This rapid voltage jump pushes displacement current ($I = C_{rss} \times dv/dt$) through Q2's internal drain-to-gate capacitance (C_{rss}) into Q2's gate. If Q2's gate resistance to ground is too high, this current creates a voltage spike that can exceed Q2's gate threshold (1.8 V min) and falsely turn Q2 back ON. Clamping the gate to 0 V through a sub-1 Ω path drains this current safely to ground.
- **Pull-Down Resistors (R4, R7 = 10 kΩ):** Placed directly between Gate and Source so that if the driver IC is unpowered, the gate is held safely at 0 V instead of floating.

2.4 DC-Bus Snubber and Series Bleeder Resistors (C8, R9, R10)

- Local Snubber / Decoupling Capacitor (C8 = 100 nF, 630 V): Placed directly across DC_PLUS and GNDPWR right next to the MOSFETs to act as a low-inductance local charge reservoir and absorb high-frequency switching transients.
- Series Bleeder Resistors (R9, R10 = 220 kΩ each): Bleeder resistors safely drain stored high voltage from the DC bus capacitors once the bench supply is turned off. Standard 0.25 W through-hole axial resistors have a maximum working voltage rating of 250 V. Placing a single resistor across 400 V would exceed its voltage rating and risk internal arcing. By wiring two 220 kΩ resistors in series ($R_{total} = 440\ k\Omega$), the voltage splits evenly:
  $V_{R9} = V_{R10} = 400\ V / 2 = 200\ V$ (Safe, < 250 V rating)
  The continuous power dissipated in each resistor at 400 V is:
  $P = V^2 / R = (200\ V)^2 / 220{,}000\ \Omega \approx 0.182\ W$ (Safe, < 0.25 W rating)

3. Two-Layer PCB Layout & High-Voltage Design Rules

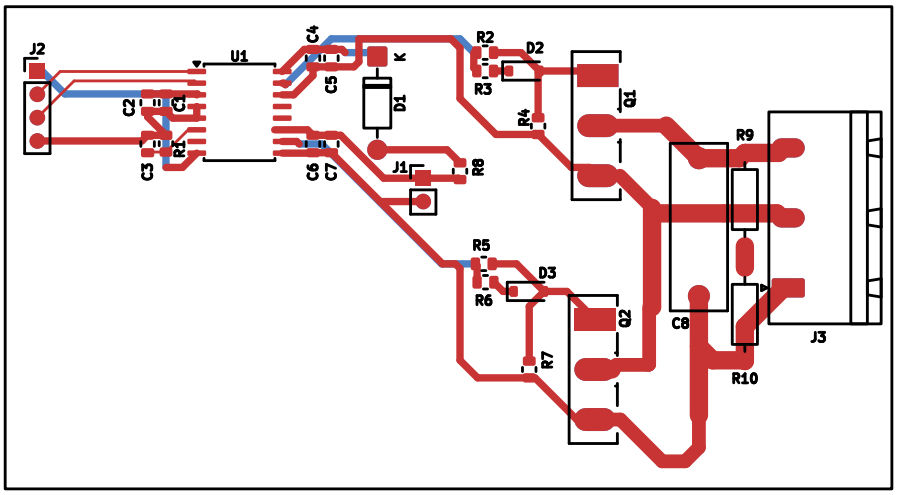


Figure 2 — Two-layer PCB layout in KiCad showing the primary logic side (left), the copper keep-out isolation barrier under U1 (center), and the 400 V power stage with necked-down TO-247 connections and compact snubber placement (right).

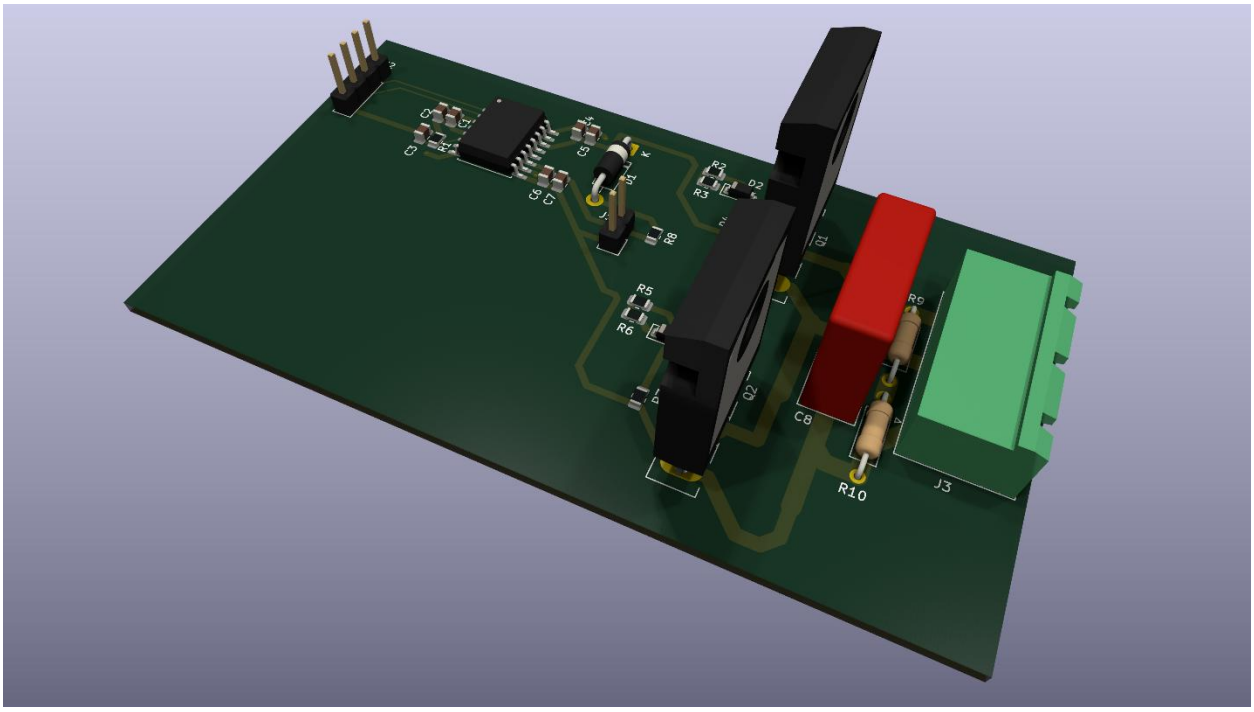


Figure 3 — 3D view of the assembled PCB verifying component spacing and mechanical clearance between the 630 V film capacitor (C8), series bleeder resistors (R9, R10), and HV terminal block (J3).

Designing a PCB for 400 V requires strict physical spacing to prevent electrical arcing across the board surface (creepage) or through the air (clearance), alongside compact routing to minimize stray inductance:

1. **Isolation Barrier Keep-Out:** No copper traces or pours are routed underneath the UCC21520DW (U1) on either the top or bottom layer, preserving the full >8 mm physical separation between the 5 V logic ground (GND) and the high-voltage domain (GNDPWR / SW_NODE).
2. **Custom 2.9 mm High-Voltage DRC Rules:** Standard IPC-2221B guidelines require at least 2.5 mm of spacing on an uncoated board at 400 V. To guarantee safety, custom Design Rule Check (DRC) rules were written in KiCad to enforce a strict 2.9 mm minimum clearance between DC_PLUS / SW_NODE and all other nets (excluding nets within the same floating 15 V high-side island, such as VDDA, OUTA, and HS_GATE). All spacing violations were resolved until the board passed with 0 DRC errors.
3. **TO-247 Pad Neck-Downs:** The physical metal pins of a vertical TO-247-3 MOSFET (Q1 and Q2) are spaced 5.45 mm apart center-to-center, leaving only ~2.6 mm between the edges of the copper pads. Running 2.5 mm wide power traces straight into the pads would dangerously narrow the gap between adjacent 400 V pins. To solve this, the DC_PLUS, SW_NODE, and GNDPWR traces were "necked down" to 1.5 mm right at the Q1 and Q2 pads and routed away at 90° angles before widening back to 2.5 mm (documenting the 2.6 mm pad-to-pad distance as a package-limited constraint).
4. **Top-Layer SW_NODE Routing & Compact Power Stage:** Originally, SW_NODE dropped to the bottom layer through a via to reach output connector J3. Because vias add stray inductance and create high-voltage spacing hazards on the bottom layer, SW_NODE was rerouted entirely on the top layer (F.Cu), passing midway between C8 Pad 1 and the 200 V R9/R10 midpoint. Furthermore, C8, R9/R10, and J3 were stacked closely beside Q1 and Q2 to keep the high-current switching loop as short as possible.

4. LTspice Simulation & Verification

4.1 LTspice Model vs. KiCad Schematic (Simplifications & Reasoning)

To verify that the gate drive network, dead-time setting, and 400 V switching stage operate properly before manufacturing hardware, the secondary side of the circuit was simulated in LTspice 26 over a 20 µs transient window (.tran 0 20u 0 1n, covering two full 100 kHz switching periods).

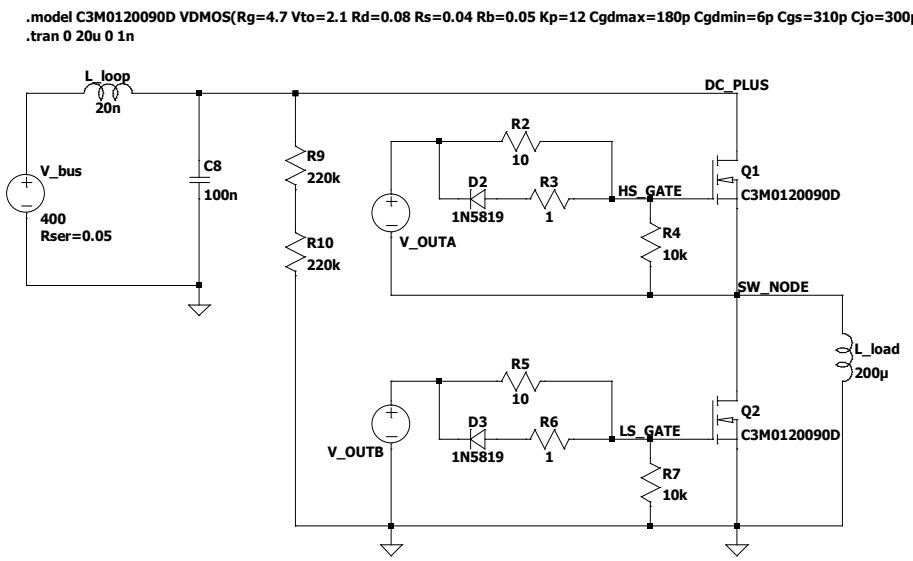


Figure 4 — LTspice transient simulation model of the secondary-side half-bridge switching cell, showing the behavioral gate driver pulse sources (V_OUTA, V_OUTB), split gate drive networks, calibrated Wolfspeed C3M0120090D VDMOS models, supply lead inductance (L_loop), snubber/bleeder network, and RL load (L_load).

Rather than copying the entire KiCad schematic exactly, several standard engineering simplifications were made in LTspice to avoid convergence bugs and model external bench connections:

1. **Replacing the UCC21520DW IC and Bootstrap Circuit with Behavioral Pulse Sources (V_OUTA and V_OUTB):**

    a. *How it differs:* In KiCad, U1 receives 5 V PWM signals and uses D1, R8, C4, and C5 to generate the high-side floating supply. In LTspice, importing vendor-encrypted gate driver IC macro-models often causes time-step convergence errors. Since the internal logic of the TI chip does not need to be redesigned, U1 and the bootstrap components are replaced by two floating PULSE voltage sources (V_OUTA and V_OUTB) connected directly to the split gate resistor networks.

    b. *How the Bootstrap Drop and Dead-Time are preserved:*

        i. V_OUTA has its negative terminal connected directly to SW_NODE (just like Pin 14 VSSA on the PCB) and pulses between 0 V and 14 V (PULSE(0 14 1u 15n 15n 4u 10u)), directly modeling the 1 V forward drop lost across bootstrap diode D1 (15 V - 1 V = 14 V).

        ii. V_OUTB is referenced to ground (0) and pulses between 15 V and 0 V (PULSE(15 0 0.78u 15n 15n 4.44u 10u)).

iii. Both sources use realistic 15 ns rise and fall times matching the UCC21520DW datasheet. V_OUTB turns OFF at 0.78 µs and V_OUTA turns ON at 1.00 µs (and V_OUTA turns OFF at 5.00 µs while V_OUTB turns ON at 5.22 µs), creating the exact 220 ns hardware dead-time programmed by R1 (22 kΩ).

2. **Adding External Bench Supply Parasitics (V_bus and L_loop) and Load (L_load):**
   a. *How it differs:* The KiCad schematic ends at screw terminal J3. To simulate actual power switching in LTspice, a 400 V DC supply (V_bus with R_ser = 0.05 Ω), power supply cable inductance (L_loop = 20 nH with R_ser = 0.05 Ω and parallel high-frequency damping R_par = 20 Ω), and an inductive load (L_load = 200 µH with series load resistance R_ser = 20 Ω) are connected to the half-bridge.
   b. *Reasoning:* Placing L_loop between V_bus and C8 models the inductance of the bench supply wires while allowing C8 (100 nF, R_ser = 50 mΩ) to decouple the MOSFET leg locally. Adding R_ser = 20 Ω to L_load represents the resistive load of the system, preventing the inductor current from ramping to infinity over multiple cycles and capping the peak load current during the 4 µs on-time at a realistic value:

   I_load,peak = (V_bus / R_load) × (1 - e^(-t_on / (L / R))) = (400 V / 20 Ω) × (1 - e^(-4 µs / 10 µs)) ≈ 20 A × (1 - e^-0.4) ≈ 6.59 A

3. **Custom Calibrated VDMOS Model for C3M0120090D:**

Instead of using a generic silicon MOSFET, a custom SPICE .model directive was defined using parameters from the Wolfspeed C3M0120090D datasheet:

.model C3M0120090D VDMOS(Rg=4.7 Vto=2.1 Rd=0.08 Rs=0.04 Rb=0.05 Kp=12 Cgdmax=180p Cgdmin=6p Cgs=310p Cjo=300p Is=1p N=6.5 Vj=4.8 m=0.5 ksubthres=0.1)

Here, Vto=2.1 sets the 2.1 V typical gate threshold, Rd=0.08 and Rs=0.04 sum to the 120 mΩ on-resistance (R_DS(on)), Rg=4.7 sets the internal gate resistance, Cgs=310p and Cgdmax=180p match the device's input and Miller capacitances, and N=6.5 / Vj=4.8 model the higher forward voltage drop of a SiC body diode.

4.2 Simulation Waveforms & Results

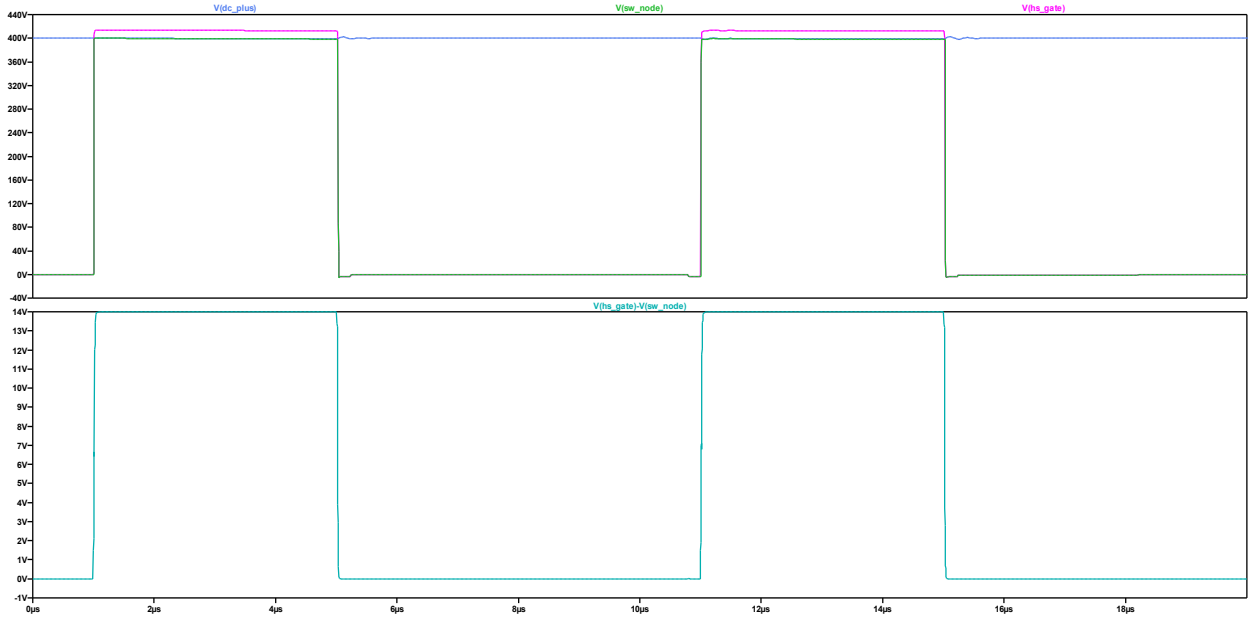


Figure 5 — Simulated waveforms over two 100 kHz switching periods (20 µs), showing stable 400 V DC bus voltage V(dc_plus), clean 0 V to 400 V switching at V(sw_node), floating high-side gate voltage V(hs_gate) reaching 414 V, and differential high-side gate-to-source drive V(HS_GATE, SW_NODE) pulsing cleanly from 0 V to 14 V.

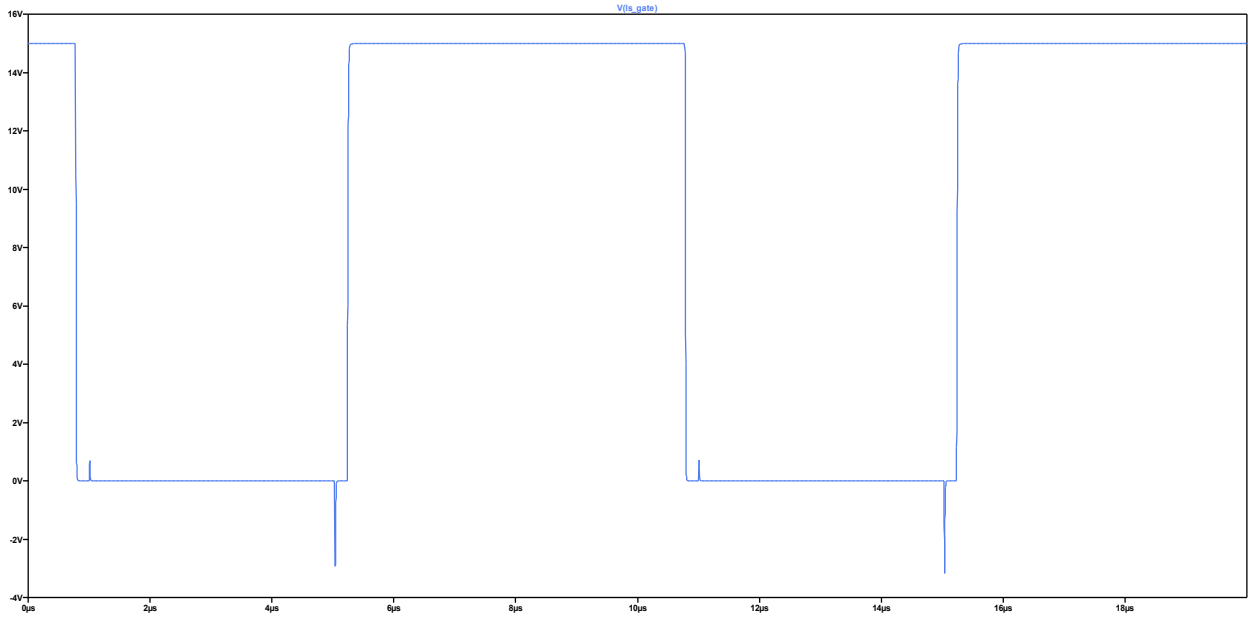


Figure 6 — Low-side gate-to-source waveform V(ls_gate) showing +15 V on-state drive, the 220 ns dead-time interval, and the small positive Miller bounce spike at t = 1.0 µs and 11.0 µs clamped safely below the 1.8 V MOSFET threshold by the 1 Ω + 1N5819 turn-off path.

Key results verified from the simulation plots include:

1. **Clean 400 V Switching & Stable DC Bus:** When Q1 turns ON at t = 1.0 µs and 11.0 µs, V(sw_node) transitions cleanly from 0 V to 400 V without shoot-through. The local 100 nF snubber (C8) absorbs the switching edge, keeping V(dc_plus) flat at 400 V with minimal, rapidly damped ringing.
2. **Verified High-Side Floating Gate Drive (14 V):** Relative to ground, V(hs_gate) shifts up to 414 V when SW_NODE is at 400 V. Plotting the differential gate-to-source voltage V(HS_GATE, SW_NODE) confirms a clean 0 V to 14 V square wave across Q1's gate and source with minimal rounded edges.
3. **Asymmetric Switching Edges & Miller Bounce Suppression:** On V(ls_gate), the rising edge charges smoothly through the 10 Ω turn-on resistor (R5), while the falling edge drops sharply to 0 V through the 1N5819 diode (D3) and 1 Ω resistor (R6). When Q1 switches on at 1.0 µs and SW_NODE slews to 400 V, the induced Miller bounce on V(ls_gate) peaks at only ~0.7 V to 0.8 V. Because this is well below the 1.8 V minimum gate threshold of the C3M0120090D, false turn-on is prevented.

5. Design Limitations Identified & Future Improvements

Through post-layout review and simulation analysis, four practical engineering trade-offs were identified that could be addressed in a newer board:

1. **Adding a Negative Turn-Off Gate Bias (-4 V / +15 V):**
   The C3M0120090D SiC MOSFET has a low gate threshold voltage (1.8 V minimum at 25°C, which drops further as the chip heats up). While our unipolar 0 V / +15 V gate drive with a 1 Ω turn-off path kept the simulated Miller spike down to ~0.8 V, real hardware PCB trace inductance in the gate loop can increase this bounce. Wolfspeed recommends driving the gate to -4 V during turn-off. In Rev 2.0, adding a small isolated bipolar DC-DC supply (-4 V / +15 V) would provide an extra 4 V of noise margin against Miller shoot-through.

2. **Compensating for the Bootstrap Diode Voltage Drop:**
   Because the bootstrap diode (D1) drops ~1.0 V, the high-side MOSFET is driven at V_GS ≈ 14 V while the low-side MOSFET receives the full 15 V. Wolfspeed specifies 15 V ± 5% (14.25 V to 15.75 V) to minimize R_DS(on). Driving Q1 at 14 V slightly increases its conduction loss compared to Q2. In Rev 2.0, replacing D1 with a lower-drop SiC Schottky diode or slightly raising the auxiliary supply voltage to 15.8 V would deliver a full 15 V to the high side.

3. **Tightening Dead-Time to Reduce Body-Diode Losses:**

During the 220 ns dead-time window when both switches are OFF, the inductive load current must freewheel through the SiC MOSFET's internal body diode. Unlike silicon diodes (~0.7 V drop), SiC body diodes have a very high forward drop (~4.2 V to 4.8 V), which wastes power as heat during dead-time. Once initial bench testing confirms clean switching without shoot-through, swapping R1 from 22 kΩ down to 10 kΩ will shorten the dead-time to ~100 ns and improve efficiency.

4. **Upgrading from a 2-Layer to a 4-Layer PCB:**
   On a 2-layer board with vertical through-hole TO-247 packages, the physical gate drive loops and main power commutation loop (C8 -> Q1 -> Q2 -> C8) have a larger loop area than ideal. Moving to a 4-layer PCB would allow placing a solid ground/return plane on an inner layer directly underneath the high-current traces, canceling magnetic flux and significantly lowering stray loop inductance.